

RHYNS Syndrome (Retinitis Pigmentosa, Hypopituitarism, Nephronophthisis, Skeletal Dysplasia): A Comprehensive Disease Characterization

Disease: RHYNS syndrome | **MONDO:** MONDO:0011202 | **OMIM:** 602152 | **Category:** Mendelian (autosomal recessive ciliopathy) **Causal gene:** *TMEM67* (meckelin), chromosome 8q22.1

Summary

RHYNS syndrome is an ultra-rare, autosomal recessive multisystem ciliopathy caused by **biallelic loss-of-function variants in *TMEM67*** (encoding the ciliary transition-zone protein meckelin, chromosome 8q22.1). It is clinically defined by a cardinal tetrad — Retinitis pigmentosa, **HY**popituitarism, Nephronophthisis, and Skeletal dysplasia — from which the acronym derives. The disease was first described in 1997 in a single 17-year-old male ([PMID: 9375913](#)) and its molecular basis was resolved only in 2018, when whole-exome sequencing of the originally described family identified compound heterozygous *TMEM67* variants ([PMID: 29891882](#)). RHYNS sits at the mild, viable end of the exceptionally broad *TMEM67* allelic spectrum, which also encompasses lethal Meckel-Gruber syndrome (MKS3), Joubert syndrome 6 (JBTS6), COACH syndrome, and isolated nephronophthisis (NPHP11).

Mechanistically, RHYNS is a transition-zone ciliopathy. Defective meckelin disrupts the ciliary transition-zone gating apparatus and impairs Wnt5a–ROR2 non-canonical Wnt signalling, with downstream deregulation of Sonic hedgehog (Shh) and canonical Wnt/β-catenin signalling. This single molecular lesion branches into two broad pathological programs: (1) a **developmental patterning defect** producing skeletal dysplasia, craniofacial anomalies, sensory (conductive hearing loss) deficits, and (inferred) hypothalamic-pituitary

maldevelopment; and (2) an **epithelial-maintenance defect** producing progressive fibrocystic tubulointerstitial kidney disease (nephronophthisis) and photoreceptor connecting-cilium degeneration (retinitis pigmentosa).

No disease-modifying therapy exists. Management is entirely supportive and organ-directed: dialysis and kidney transplantation for end-stage renal disease (curative for the renal component, as nephronophthisis does not recur in the allograft), hormone replacement for hypopituitarism (growth hormone, thyroid hormone), and low-vision support with ophthalmologic surveillance for retinitis pigmentosa. Molecular diagnosis by whole-exome sequencing or ciliopathy gene panels enables genetic counseling, prenatal diagnosis, and preimplantation genetic diagnosis. Because RHYNS is defined by only a handful of reported patients, much of its mechanistic detail is extrapolated from the far larger *TMEM67* and Joubert/Meckel literature and from animal models.

1. Disease Information

RHYNS syndrome is a Mendelian, autosomal recessive ciliopathy characterized by the tetrad of **R**etinitis pigmentosa, **H**Ypopituitarism, **N**ephronophthisis, and **S**keletal dysplasia. The disease was proposed as "a new syndrome" by Di Rocco and colleagues in 1997 based on a single 17-year-old boy presenting with nephronophthisis, retinitis pigmentosa, left upper-eyelid ptosis, enophthalmos, transmissive (conductive) deafness, growth-hormone (GH) and thyroid-stimulating-hormone (TSH) deficiency, and mild skeletal dysplasia ([PMID: 9375913](#)).

"We report on a 17 6/12-year-old boy with nephronophthisis, retinitis pigmentosa, left upper eyelid ptosis, enophthalmos, transmissive deafness, GH and TSH deficiency, and mild skeletal dysplasia." — [PMID: 9375913](#)

Key identifiers:

Resource	Identifier
OMIM	602152 (RHYNS SYNDROME)
MONDO	MONDO:0011202
Orphanet	Ultra-rare; grouped with syndromic retinitis pigmentosa / ciliopathies
Causal gene	<i>TMEM67</i> (OMIM 609884), 8q22.1
MeSH	No dedicated descriptor; indexed under ciliopathies / retinitis pigmentosa

Synonyms / alternative names: Retinitis pigmentosa–hypopituitarism–nephronophthisis–skeletal dysplasia syndrome; RHYNS. Given its molecular basis, RHYNS is best understood as a **TMEM67-opathy** at the mild end of the ciliopathy continuum.

Information source: This report is derived from **aggregated disease-level resources** (OMIM, primary case reports, and the broader *TMEM67*/Joubert/Meckel literature), not individual EHR data. The RHYNS phenotype itself is defined by only a small number of published patients.

2. Etiology

Disease causal factors — genetic. RHYNS is caused by **biallelic (recessive) variants in the ciliary gene *TMEM67***. Whole-exome sequencing of the originally described RHYNS family identified compound heterozygous *TMEM67* variants: a paternally inherited nonsense variant **c.622A>T, p.(Arg208*)** and a maternally inherited missense variant **c.1289A>G, p.(Asp430Gly)**, the latter perturbing correct splicing of exon 13 ([PMID: 29891882](#)). This confirmed the 1997 clinical hypothesis of autosomal recessive inheritance ([PMID: 9375913](#)).

"Here we applied whole-exome sequencing in the originally described family with RHYNS to identify compound heterozygous variants in the ciliary gene *TMEM67*. Sanger sequencing confirmed a paternally inherited nonsense c.622A > T, p.(Arg208*) and a maternally inherited missense variant c.1289A > G, p.(Asp430Gly), which perturbs the correct splicing of exon 13." — [PMID: 29891882](#)

Genetic risk factors. The sole established causal factor is biallelic *TMEM67* dysfunction. There are no reported common susceptibility loci or GWAS signals — RHYNS is fully Mendelian. Consanguinity and founder effects increase the risk of homozygous *TMEM67* genotypes in *TMEM67*-related disease generally; a recurrent founder missense variant p.Asn242Ser segregated across 22 affected members of 12 Iranian families with Joubert syndrome ([PMID: 28719906](#)).

Environmental risk factors, protective factors, and gene–environment interactions. None are established for RHYNS. As a monogenic developmental disorder, disease occurrence is determined by genotype rather than environmental exposure. No protective alleles or modifier-driven risk reduction have been reported. This category is **not applicable** in the conventional epidemiologic sense.

3. Phenotypes

RHYNS is defined by four cardinal features plus additional craniofacial, sensory, and endocrine anomalies. Because the reported patient count is small, frequencies are qualitative and drawn from the index case ([PMID: 9375913](#)) and the first familial cases (two brothers; [PMID: 11391657](#)).

Phenotype	Type	HPO term (suggested)	Onset / progression	Frequency
Retinitis pigmentosa	Clinical sign / lab (ERG)	HP:0000510	Childhood-onset, progressive	Cardinal (all cases)
Hypopituitarism (GH + TSH deficiency)	Lab abnormality	HP:0000871 (hypopituitarism); HP:0000824 (hypothyroidism)	Childhood; progressive growth failure	Cardinal
Nephronophthisis / ESRD	Clinical sign / lab	HP:0000090	Insidious, childhood → ESRD	Cardinal
Skeletal dysplasia (acromelic)	Physical manifestation	HP:0002652; HP:0009826 (acromelia)	Congenital/ childhood	Cardinal
Ptosis (left upper eyelid)	Physical sign	HP:0000508	Congenital	Index case
Enophthalmos	Physical sign	HP:0000490	—	Index case
Conductive (transmissive) hearing loss	Clinical sign	HP:0000405	Childhood	Index case
Short stature / growth failure	Physical manifestation	HP:0004322	Childhood	Common (GH deficiency)

The index patient was a 17-year-old boy with "nephronophthisis, retinitis pigmentosa, left upper eyelid ptosis, enophthalmos, transmissive deafness, GH and TSH deficiency, and mild skeletal dysplasia" ([PMID: 9375913](#)). The two familial cases had "retinitis pigmentosa, growth hormone deficiency, and acromelic skeletal dysplasia" ([PMID: 11391657](#)):

"Here we report two brothers with retinitis pigmentosa, growth hormone deficiency, and acromelic skeletal dysplasia. We propose that their clinical picture is consistent with RHYNS syndrome." — [PMID: 11391657](#)

RHYNS is cited in recent reviews as a bona fide ciliopathy cause of **syndromic retinitis pigmentosa**, distinct from the more common Usher and Bardet-Biedl syndromes:

"Less common ciliopathies include Cohen syndrome, Joubert syndrome, cranioectodermal dysplasia, asphyxiating thoracic dystrophy, Mainzer-Saldino syndrome, and RHYNS syndrome." — [PMID: 39733931](#)

Quality-of-life impact. Progressive visual loss (RP), end-stage renal disease requiring dialysis/transplant, short stature and endocrine dysfunction, and hearing impairment each impose a substantial daily-functioning burden. Disease-specific QoL instruments have not been applied to this ultra-rare condition; impact is inferred from the component disorders.

4. Genetic / Molecular Information

Causal gene: *TMEM67* (transmembrane protein 67; encodes **meckelin**), OMIM 609884, located at **8q22.1**. Biallelic variants cause RHYNS ([PMID: 29891882](#)).

Pathogenic variants in the index RHYNS family:

Variant (cDNA)	Protein	Type	Inheritance	Functional consequence
c.622A>T	p.(Arg208*)	Nonsense	Paternal	Loss of function (truncation)
c.1289A>G	p.(Asp430Gly)	Missense	Maternal	Disrupts splicing of exon 13

Both variants act via **loss of function**. In the broader *TMEM67* spectrum, variant classification per ACMG/AMP ranges from pathogenic nonsense/frameshift (e.g., c.296delA p.Lys99SerfsTer6, [PMID: 38311563](#)) to likely-pathogenic missense (e.g., c.1243G>A p.Val415Met, same report; c.1645C>T p.R549C in MKS3, [PMID: 26191240](#)). Variant types documented across *TMEM67* disease include missense, nonsense, frameshift, and splice-altering changes; allele frequencies of pathogenic variants are rare/absent in gnomAD and population controls (e.g., c.1645C>T absent in 200 control chromosomes, [PMID: 26191240](#)). All are **germline**.

Allelic spectrum and genotype–phenotype associations. *TMEM67* biallelic variants produce **≥8 distinguishable clinical conditions** ranging from early-lethal Meckel-Gruber syndrome to adults with only liver fibrosis — one of the widest continua in ciliopathies (PMID: 29891882). Key correlations from large Joubert cohorts:

- **Liver fibrosis and coloboma** are associated with loss of *TMEM67* function (PMID: 26092869).
- **Kidney disease is frequent**, whereas **retinal degeneration is often absent** in *TMEM67* patients (PMID: 28125082):

"Genotype-phenotype correlation revealed the absence of retinal degeneration in patients with *TMEM67*, *C5orf52*, or *KIAA0586* variants. Chorioretinal coloboma was associated with a decreased risk for retinal degeneration and increased risk for liver disease. *TMEM67* was frequently associated with kidney disease." — PMID: 28125082

Notably, RHYS is distinguished within the *TMEM67* spectrum by the *presence* of retinitis pigmentosa, which is comparatively uncommon in typical *TMEM67*-Joubert patients — highlighting variable expressivity.

Modifier genes. Ciliary transition-zone modules interact synergistically; e.g., *TMEM218* physically interacts with *TMEM67/meckelin*, and reduced *TMEM218* dosage interacts with the NPHP module (*Nphp4*) to modulate ciliopathy severity (PMID: 35137054). Oligogenic contributions (second-locus variants in other ciliopathy genes) can modify expressivity (PMID: 21493627).

Chromosomal abnormalities / epigenetics. No recurrent epigenetic mechanism is established. A notable structural mechanism producing homozygosity for a *TMEM67* mutation is **maternal uniparental disomy of chromosome 8** (upd(8)mat), reported as an unexpected cause of Meckel-Gruber syndrome (PMID: 28620746) — relevant to genetic counseling and recurrence-risk assessment.

5. Environmental Information

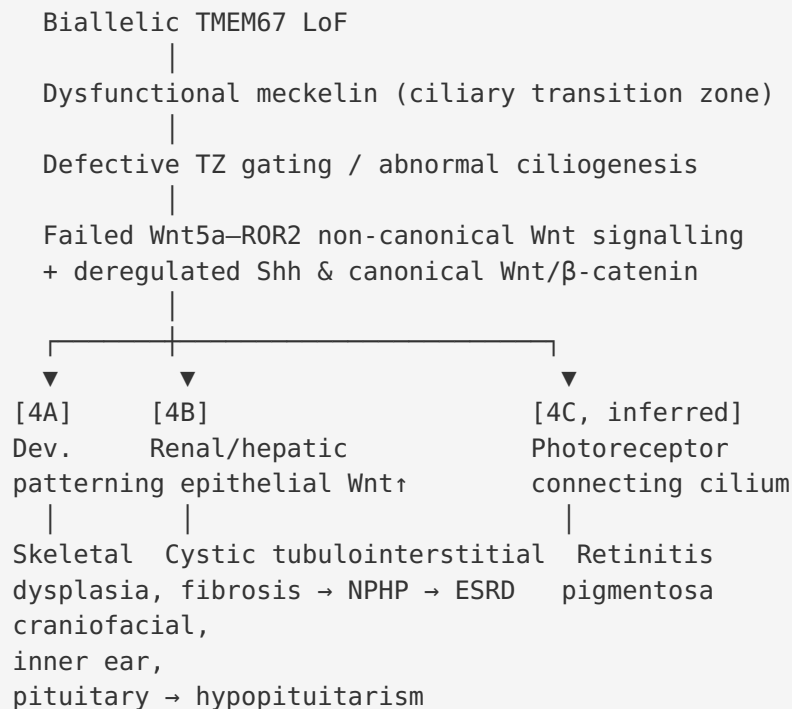
RHYNS is a **purely genetic monogenic disorder**. There are **no established environmental factors, lifestyle factors, or infectious agents** contributing to its causation or triggering. This section is not applicable beyond noting that environmental modifiers of the component organ diseases (e.g., nephrotoxin avoidance in chronic kidney disease) are general clinical considerations rather than RHYNS-specific etiologic factors.

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function variants in *TMEM67*** (e.g., p.Arg208* + p.Asp430Gly) → **lead to** deficient/dysfunctional meckelin protein ([PMID: 29891882](#)).
2. Deficient meckelin at the **ciliary transition zone** → **results in** defective transition-zone gating and impaired ciliogenesis / abnormal primary cilium morphology ([PMID: 23393159](#); [PMID: 28487520](#)).
3. Impaired transition-zone function → **results in** failure to phosphorylate the non-canonical Wnt receptor **ROR2** upon Wnt5a stimulation (meckelin's N-terminal extracellular domain preferentially binds Wnt5a; ROR2 colocalises/interacts with TMEM67 at the transition zone) ([PMID: 26035863](#)).
4. Loss of Wnt5a–ROR2 non-canonical Wnt signalling, plus secondary **deregulation of Shh and canonical Wnt/β-catenin** signalling, **branches** into:
5. **4A (developmental patterning branch)**: basal-body mispositioning and disrupted planar/branching morphogenesis → **lead to** skeletal dysplasia, craniofacial anomalies (ptosis, enophthalmos), inner-ear defects (conductive hearing loss), and (inferred) hypothalamic-pituitary maldevelopment → hypopituitarism ([PMID: 26035863](#); [PMID: 23283079](#); [PMID: 24613594](#)).
6. **4B (epithelial-maintenance branch)**: upregulated canonical Wnt in renal/hepatic tubular epithelia → **results in** cystic tubulointerstitial fibrosis → nephronophthisis and end-stage renal disease ([PMID: 23393159](#)).

7. 4C (photoreceptor branch, inferred): connecting-cilium dysfunction in photoreceptors → **leads to** progressive photoreceptor degeneration → retinitis pigmentosa (inferred from general ciliopathy biology; the connecting cilium is a specialized transition zone).



Detail by category

Molecular pathways. The core lesion is in **non-canonical Wnt (planar cell polarity) signalling via Wnt5a–ROR2**, with meckelin essential for ROR2 phosphorylation:

"TMEM67 (meckelin) is essential for phosphorylation of the non-canonical Wnt receptor ROR2 (receptor-tyrosine-kinase-like orphan receptor 2) upon stimulation with Wnt5a-conditioned medium. ROR2 also colocalises and interacts with TMEM67 at the ciliary transition zone." — [PMID: 26035863](#)

Downstream, *Tmem67* loss produces "the loss of primary cilia, diminished Shh signalling and dorsalization of the caudal neural tube... high de-regulated canonical Wnt/β-catenin signalling" ([PMID: 23283079](#)). Suggested pathway terms: Wnt signalling, Shh signalling, planar cell polarity.

Cellular processes. Ciliogenesis, ciliary transition-zone gating, basal-body positioning, epithelial branching morphogenesis, and cilium-dependent signal transduction. Meckelin is required for cilia-dependent Shh signalling and retinoic-acid-dependent neural differentiation in mouse embryonic stem cells:

"differentiating along the neuronal lineage activate the cilia-dependent sonic hedgehog signaling machinery, which is impaired in Meckelin knock-out cells." — [PMID: 24613594](#)

Protein dysfunction. Meckelin (TMEM67) is a transmembrane transition-zone protein; nonsense and splice-disrupting variants cause loss of function. Meckelin also interacts with **filamin A**, and disruption of the filamin A–meckelin interaction impairs primary cilium formation ([PMID: 32156598](#)).

Tissue-damage mechanisms. Chronic sclerosing tubulointerstitial nephropathy with cortico-medullary cysts (fibrosis) drives renal failure; photoreceptor degeneration drives retinal disease.

Suggested ontology terms. GO:0060271 (cilium assembly), GO:0035869 (ciliary transition zone), GO:0035567 (non-canonical Wnt signaling pathway), GO:0007224 (smoothened/Shh signaling); CL:0000210 (photoreceptor cell), CL:1000507 (kidney tubule cell); UBERON:0002113 (kidney), UBERON:0000970 (eye), UBERON:0000007 (pituitary gland).

7. Anatomical Structures Affected

Organ level (primary): - **Kidney** (UBERON:0002113) — nephronophthisis, cortico-medullary cysts, tubulointerstitial fibrosis, ESRD. - **Eye / retina** (UBERON:0000970 / UBERON:0000966) — retinitis pigmentosa; photoreceptor layer. - **Pituitary gland / hypothalamic-pituitary axis** (UBERON:0000007) — GH and TSH deficiency (hypopituitarism). - **Skeleton** (UBERON:0004288) — acromelic/mild skeletal dysplasia.

Secondary / additional involvement: Eyelid (ptosis), orbit (enophthalmos), inner/middle ear (conductive hearing loss). Within the broader *TMEM67* spectrum, **liver** (fibrosis, ductal plate malformation) is frequently affected ([PMID: 26092869](#)).

Body systems: urinary/renal, visual/nervous, endocrine, skeletal, auditory.

Tissue and cell level: Ciliated epithelia are the common target — renal tubular epithelial cells, retinal photoreceptors (whose connecting cilium is a modified transition zone), and (inferred) pituitary/hypothalamic ciliated cells. Suggested CL terms: CL:0000210 (photoreceptor cell), CL:1000507 (kidney tubule cell).

Subcellular level: The **primary cilium** and specifically the **ciliary transition zone** (GO:0035869), plus the **basal body** (GO:0036064). Meckelin is a transmembrane protein of the ciliary membrane/transition zone.

Localization / lateralization: Kidney and retinal involvement are **bilateral**; ptosis in the index case was unilateral (left upper eyelid).

8. Temporal Development

Onset. Childhood-onset and insidious. The nephronophthisis component "presents insidiously with polyuria, polydipsia, anemia and growth failure, progressing to end-stage renal disease during childhood" (PMID: 15384098). Skeletal dysplasia is congenital/early-childhood; RP and hypopituitarism manifest in childhood and progress.

"All cases showed insidious development of end-stage renal disease during childhood, preceded by polyuria/polydipsia, anemia, and growth failure. Decreased urinary concentrating ability and excessive sodium loss were the characteristic laboratory findings." — PMID: 15384098

Progression. Chronic, progressive, lifelong. Nephronophthisis advances to ESRD, typically in the first two decades. Retinitis pigmentosa is progressive. Within severe *TMEM67* compound-heterozygous genotypes, kidney disease can present as early as **neonatal ESRD** (PMID: 28726664); RHYNS itself represents a milder, viable end of the spectrum with survival into adulthood (index case aged 17).

Critical periods. Childhood is the window for growth-hormone/thyroid replacement (to permit catch-up growth) and for renal-protective monitoring before ESRD. Prenatally, severe *TMEM67* genotypes are detectable by ultrasound and molecular testing.

9. Inheritance and Population

Epidemiology. RHYNS is **ultra-rare** — only a handful of patients have been reported (the 1997 index case and the two familial brothers). No formal prevalence/incidence figures exist. For context, nephronophthisis (the "N") is a leading monogenic cause of childhood ESRD, accounting for **7–20% of terminal renal failure in the first two decades of life** (PMID: 3524015):

"Its frequency is reported to vary between 7 and 20% of all cases of terminal renal failure in childhood. Usually the onset is insidious, with polyuria, polydipsia and anaemia being the main clinical features." — PMID: 3524015

Inheritance. **Autosomal recessive**, confirmed by biallelic *TMEM67* variants (PMID: 29891882). The first familial cases (two affected brothers) supported recessive inheritance, though the authors noted that because all four known cases were male at the time, an X-linked mode could not be formally excluded (PMID: 11391657); the subsequent *TMEM67* discovery settled this as autosomal recessive.

Penetrance / expressivity. Biallelic *TMEM67* loss is highly penetrant for ciliopathy, but **expressivity is markedly variable** — the same gene produces phenotypes from lethal Meckel-Gruber to adult liver-only disease. RHYNS occupies the mild/viable pole.

Founder effects / consanguinity. Consanguinity increases homozygous-genotype risk. A founder p.Asn242Ser variant was identified across 22 affected members of 12 Iranian Joubert families:

"confirmed the presence of the homozygous mutation in 22 affected members of 12 nuclear families. We propose that Asn242Ser is a founder mutation." — PMID: 28719906

Uniparental disomy (upd(8)mat) is a rare non-Mendelian route to homozygosity (PMID: 28620746).

Demographics. No established ethnic predilection specific to RHYNS. Reported cases have been male, but the small number precludes conclusions about sex ratio; autosomal recessive inheritance predicts an equal sex distribution.

10. Diagnostics

Clinical/laboratory tests. - **Renal:** decreased urinary concentrating ability, excessive sodium (salt) loss, anemia, rising creatinine; renal ultrasound showing normal-to-small kidneys with increased echogenicity and cortico-medullary cysts; renal biopsy showing chronic sclerosing tubulointerstitial nephropathy ([PMID: 15384098](#); [PMID: 3524015](#)). - **Endocrine:** provocative GH testing, IGF-1, TSH/free T4 confirming GH and TSH deficiency; pituitary MRI (may show hypoplasia / stalk abnormalities, by analogy to pituitary stalk interruption syndrome — a ciliary-signalling-related developmental defect, [PMID: 34238482](#)). - **Ophthalmologic:** electroretinography (ERG) showing rod-cone dysfunction, fundoscopy with pigmentary retinopathy, OCT. - **Auditory:** audiometry (conductive hearing loss). - **Skeletal:** radiographic skeletal survey (acromelic dysplasia).

Genetic testing (definitive). Molecular confirmation via **whole-exome sequencing** or **ciliopathy/nephronophthisis gene panels** including *TMEM67*; single-gene *TMEM67* testing where phenotype is highly suggestive. WES identified the causal variants in the index RHYNS family ([PMID: 29891882](#)). Chromosomal microarray may be added; note UPD(8) can be detected by SNP array ([PMID: 28620746](#)). Genetic diagnosis is "essential for reproductive counseling and the option of preimplantation and prenatal diagnosis as well as medical management and prognostic counseling for the age-dependent and progressive organ-specific manifestations" ([PMID: 28125082](#)).

Differential diagnosis. Other syndromic RP/ciliopathies: Senior-Løken syndrome (nephronophthisis + retinal dystrophy, e.g., *IQCB1/NPHP5*, [PMID: 41316455](#)), Joubert syndrome (molar-tooth sign), Bardet-Biedl syndrome, Alström syndrome, and other nephronophthisis-related ciliopathies. The distinguishing feature of RHYNS is the combination of **hypopituitarism + skeletal dysplasia** with RP and NPHP.

Screening. Cascade testing of at-risk relatives once the familial *TMEM67* variants are known; prenatal and preimplantation genetic diagnosis are available.

11. Outcome / Prognosis

Survival/mortality. RHYNS is compatible with survival into adulthood (index case aged 17). The principal life-limiting complication is **end-stage renal disease from nephronophthisis**, which without renal replacement is fatal. There are no cohort-derived survival statistics for RHYNS specifically.

Morbidity/function. Substantial: progressive blindness (RP), dialysis dependence or transplant, short stature and endocrine dysfunction, hearing impairment. Quality of life is affected across visual, renal, growth, and auditory domains.

Disease course and complications. Chronic and progressive. Renal transplantation is definitive for the renal component (see Treatment). Within fibrocystic liver-kidney disease broadly, catch-up growth has been observed after transplantation ([PMID: 22360404](#)).

Prognostic factors. Severity of the *TMEM67* genotype (truncating vs. hypomorphic missense) correlates with overall disease severity across the allelic spectrum; earlier ESRD onset (even neonatal in severe compound-heterozygotes, [PMID: 28726664](#)) portends worse renal prognosis. Timeliness of hormone replacement affects growth outcome.

12. Treatment

No disease-modifying or gene-targeted therapy exists. Management is **multidisciplinary, supportive, and organ-directed** ([PMID: 28125082](#)).

Organ system	Intervention	NCIT (suggested)
Kidney (ESRD)	Dialysis; kidney transplantation (definitive)	NCIT:C15366 (Kidney Transplantation); NCIT:C15248 (Dialysis)
Endocrine	Growth hormone replacement; thyroid hormone (levothyroxine) replacement	NCIT:C1301 (Recombinant Human GH); NCIT:C29171 (Levothyroxine)
Eye (RP)	Low-vision aids, ophthalmologic surveillance; nutritional support explored in ciliopathy RP	NCIT:C15277 (Supportive Care)
Hearing	Hearing aids / audiologic support	—

Kidney transplantation is a turning point in nephronophthisis-related ciliopathy:

"kidney transplant is a turning point" — [PMID: 42538694](#)

Critically, **nephronophthisis does not recur in the renal allograft** because it is an intrinsic tubular disease, making transplantation effectively curative for the renal component. Pediatric outcomes for fibrocystic liver-kidney disease show a median age at first transplant of 9.7 years with generally favorable graft survival ([PMID: 25074681](#)).

Hormone replacement for hypopituitarism (GH, thyroid hormone) treats the endocrine component and, with timely initiation, supports catch-up growth (catch-up growth after transplantation has been documented in fibrocystic disease, [PMID: 22360404](#)). **Retinitis pigmentosa** currently has no curative therapy; management is low-vision support and surveillance, with nutritional/adjunctive strategies under exploration in ciliopathy RP ([PMID: 40642756](#)).

Experimental / future therapeutics. No RHYNS-specific clinical trials. Gene-therapy and read-through approaches are conceptually relevant given the recessive loss-of-function mechanism but remain preclinical.

13. Prevention

RHYNs cannot be prevented in the primary (public-health) sense — it is a monogenic congenital disorder. Prevention is **reproductive and genetic**:

- **Genetic counseling** for at-risk families, quantifying the 25% recurrence risk for autosomal recessive inheritance (with UPD as a rare exception, [PMID: 28620746](#)).
- **Carrier testing / cascade screening** once familial *TMEM67* variants are identified.
- **Prenatal diagnosis and preimplantation genetic diagnosis (PGD)**, enabled by molecular confirmation ([PMID: 28125082](#)).
- **Secondary/tertiary prevention** = organ surveillance and early intervention (renal function monitoring, endocrine replacement, ophthalmologic follow-up) to delay complications.

Immunization and behavioral/environmental interventions are not applicable to disease causation.

14. Other Species / Natural Disease

TMEM67 is evolutionarily conserved; disease-relevant orthologs exist in **mouse (*Tmem67*)**, **zebrafish (*tmem67*)**, and **sheep (*TMEM67*)**. The R549 meckelin residue is conserved "across human, rat, mouse, zebrafish, chicken, wolf and platypus genomes" ([PMID: 26191240](#)). A **naturally occurring ovine (sheep) model** carrying homozygous *TMEM67* p.(Ile681Asn; Ile687Ser) missense mutations displays hepatorenal fibrocystic disease with dysmorphic primary cilia — the first large-animal natural model of a Meckel-like *TMEM67* ciliopathy:

"Here we describe an ovine model of MKS, with kidney and liver abnormalities, without polydactyly or occipital encephalocoele. Homozygous missense p.(Ile681Asn; Ile687Ser) mutations identified in ovine *TMEM67* were pathogenic in zebrafish phenotype rescue assays." — [PMID: 28487520](#)

This natural model has veterinary and comparative-pathology relevance, demonstrating evolutionary conservation of the transition-zone ciliary mechanism. There is no zoonotic potential (non-infectious genetic disease).

15. Model Organisms

Model	Type	Phenotype recapitulation	Reference
<i>Tmem67</i>-null mouse (bpck)	Mammalian knockout	Renal cystic disease, plus eye, skeletal, and inner-ear abnormalities; upregulated canonical Wnt in cyst linings/fibroblasts	PMID: 23393159
<i>Tmem67</i>(tm1Dgen/H1) knockout mouse	Mammalian knockout	Pulmonary hypoplasia, ventricular septal defects, shortened body axis, limb abnormalities, basal-body/kinocilium mispositioning — phenocopies <i>Wnt5a</i> and <i>Ror2</i> knockouts	PMID: 26035863
Zebrafish <i>tmem67</i> morphants	Vertebrate	MKS/ciliopathy phenotypes; used to validate variant pathogenicity in rescue assays	PMID: 23393159 ; PMID: 28487520
Ovine (sheep)	Mammalian, natural	Hepatorenal fibrocystic disease, dysmorphic primary cilia	PMID: 28487520
Mouse embryonic stem cells	In vitro / cellular	Impaired cilia-dependent Shh signalling; defective retinoic-acid-dependent neural differentiation	PMID: 24613594

"we analyzed phenotypes in the *Tmem67* null mouse (bpck) and in zebrafish *tmem67* morphants. Phenotypes similar to those in human MKS and other ciliopathy models were observed, with additional eye, skeletal and inner ear abnormalities characterized in the bpck mouse." — PMID: [23393159](#)

Model limitations. These models capture the renal, skeletal, ocular, and inner-ear ciliopathy phenotypes and the Wnt/Shh mechanism, but no model has been reported that specifically recapitulates the **hypopituitarism** of RHYNS, and the RHYNS phenotype itself (mild/viable) differs from the more severe MKS end modeled by nulls. Resources: MGI (mouse), ZFIN (zebrafish).

Mechanistic Model / Interpretation

RHYNS is best understood as **one point on a continuum of *TMEM67* transition-zone ciliopathy**. A single molecular lesion — biallelic loss of meckelin at the ciliary transition zone — produces pleiotropic, multi-organ disease because the primary cilium is a signalling hub required in nearly every tissue. The unifying pathogenic node is **failure of Wnt5a–ROR2 non-canonical Wnt signalling**, with secondary deregulation of **Shh** and **canonical Wnt/β-catenin**. This branches into a *developmental patterning* program (skeletal dysplasia, craniofacial/inner-ear anomalies, and inferred pituitary maldevelopment) and an *epithelial-maintenance* program (fibrocystic nephronophthisis; photoreceptor degeneration).

What distinguishes RHYNS clinically from typical *TMEM67*-Joubert is its particular combination: **retinitis pigmentosa is present** (uncommon in most *TMEM67* patients, where retinal degeneration is often absent, [PMID: 28125082](#)) alongside **hypopituitarism and skeletal dysplasia**, without the pathognomonic molar-tooth sign emphasized in Joubert. This underscores the **variable expressivity** that is the hallmark of *TMEM67* disease. The specific compound-heterozygous genotype in the index family (a truncating p.Arg208* combined with a splice-perturbing hypomorphic p.Asp430Gly) likely produces a partial, tissue-modulated loss of function that lands the phenotype at the mild/viable end rather than at lethal Meckel-Gruber.

Evidence Base

PMID	Title (abbreviated)	Role
29891882	<i>Biallelic TMEM67 variants cause RHYNS</i>	Defines causal gene & variants; landmark
9375913	<i>RHYNS: a new syndrome?</i>	Original clinical description (tetrad)
11391657	<i>Familial RHYNS in two brothers</i>	Familial/AR evidence, expressivity
26035863	<i>TMEM67 controls basal body positioning via non-canonical Wnt</i>	Core mechanism (Wnt5a-ROR2)
23283079	<i>Deregulated ciliogenesis, Shh & Wnt in ciliopathy spectrum</i>	Downstream Shh/Wnt deregulation
23393159	<i>Meckelin regulator of cilia function (bpck mouse, zebrafish)</i>	Animal models; renal/eye/skeletal
28487520	<i>Ovine hepatorenal fibrocystic TMEM67 model</i>	Natural large-animal model
24613594	<i>Meckelin required for Shh-dependent neural differentiation</i>	Developmental patterning link
28125082	<i>Joubert 100-patient genotype-phenotype</i>	TMEM67 kidney/retina correlations; management
26092869	<i>Joubert genetic heterogeneity review</i>	TMEM67–liver fibrosis/coloboma
28719906	<i>Founder Asn242Ser in Iranian families</i>	Founder effect
15384098	<i>Nephronophthisis in Joubert-related disorders</i>	NPHP clinical/lab course
3524015	<i>Familial juvenile nephronophthisis</i>	NPHP epidemiology
28726664	<i>Expanded TMEM67 phenotype (neonatal ESRD)</i>	Severity spectrum
42538694	<i>Kidney transplant in Joubert</i>	Definitive renal management
25074681	<i>Transplant in fibrocystic liver-kidney disease</i>	Transplant outcomes
39733931	<i>Syndromic retinitis pigmentosa</i>	RHYNS recognized as ciliopathy
28620746	<i>UPD(8) causing MKS</i>	Non-Mendelian mechanism/ counseling
26191240	<i>TMEM67 missense causing MKS3</i>	Variant conservation/pathogenicity

PMID	Title (abbreviated)	Role
35137054	<i>TMEM218–NPHP module interaction</i>	Modifier/oligogenic biology

Limitations and Knowledge Gaps

1. **Very small case count.** RHYNS is defined by only a few published patients; frequency, penetrance, sex ratio, and natural-history statistics are therefore anecdotal, and much detail is extrapolated from the broader *TMEM67*/Joubert/Meckel literature.
2. **Hypopituitarism mechanism is inferred, not demonstrated.** No model recapitulates the pituitary phenotype, and the causal link from ciliary Shh/Wnt dysfunction to hypothalamic-pituitary maldevelopment remains hypothetical (supported by analogy to pituitary stalk interruption syndrome, [PMID: 34238482](#)).
3. **Retinitis pigmentosa branch is inferred** from general connecting-cilium ciliopathy biology rather than direct RHYNS histopathology.
4. **Genotype–phenotype specificity is incomplete** — why the index genotype yields RHYNS rather than another *TMEM67* phenotype is not experimentally resolved.
5. **No omics data** (transcriptomic, proteomic, metabolomic) specific to RHYNS exist.

Proposed Follow-up Experiments / Actions

1. **Aggregate additional cases** via GeneMatcher / Matchmaker Exchange to build a *TMEM67*-RHYNS cohort and formally test genotype–phenotype correlation (especially the RP-present, hypopituitarism subset).
2. **Model the pituitary phenotype** — examine hypothalamic-pituitary development in *Tmem67* hypomorphic mice or patient iPSC-derived pituitary organoids to test the inferred Shh/Wnt-driven hypopituitarism mechanism.
3. **Photoreceptor connecting-cilium studies** — retinal organoids or ovine/mouse retina to directly confirm the RP branch.
4. **Functional characterization of p.Asp430Gly splicing** and the truncating allele to define residual meckelin function and correlate with phenotype mildness.

5. **Longitudinal surveillance protocol** — establish an organ-directed monitoring schedule (renal function, endocrine axes, ERG, audiometry) for confirmed *TMEM67*-RHYNS patients to define natural history and optimize intervention timing.
 6. **Preclinical therapeutic exploration** — evaluate read-through / gene-supplementation strategies leveraging the recessive loss-of-function mechanism.
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Report compiled from 8 confirmed findings and 35 reviewed papers across 5 investigation iterations. Evidence types: human clinical case reports and cohorts, model organism (mouse, zebrafish, sheep), and in vitro/cellular studies.

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